Exercise 13.1

1. Arshad rolls a dice, with sides labelled $\boldsymbol{L, M, N, O, P, U}$. What is the probability that the dice lands on a consonant?

When a dice is rolled, the sample space will be:

$$S = \{L, M, N, O, P, U\}$$
$$n(S) = 6$$

Let A be an event getting consonant

$$A = \{L, M, N, P\}$$
$$n(A) = 4$$
$$P(A) = \frac{n(A)}{n(S)}$$
$$P(A) = \frac{4}{6}$$
$$\boldsymbol{P(A) = \frac{2}{3}}$$

2. Shazia throws a pair of fair dice. What will be the probability of getting:
(i) Sum of dots is at least 4.
(ii) Product of both dots is between 5 to 10.
(iii) The difference between both the dots is equal to 4.
(iv) Number at least 5 on the first dice and number at least 4 on the second dice.

When two dice are thrown, the sample space will be:

$$\begin{aligned} S = \{&(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),\\ &(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),\\ &(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),\\ &(4,1),(4,2),(4,3),(4,4),(4,5),(4,6),\\ &(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),\\ &(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}\end{aligned}$$
$$n(S) = 36$$

(i) Let A be the event getting the sum is at least 4.

$$\begin{aligned} A = \{&(1,3),(1,4),(1,5),(1,6),(2,2),(2,3),\\ &(2,4),(2,5),(2,6),(3,1),(3,2),(3,3),\\ &(3,4),(3,5),(3,6),(4,1),(4,2),(4,3),\\ &(4,4),(4,5),(4,6),(5,1),(5,2),(5,3),\\ &(5,4),(5,5),(5,6),(6,1),(6,2),(6,3),\\ &(6,4),(6,5),(6,6)\}\end{aligned}$$
$$n(A) = 33$$
$$P(A) = \frac{n(A)}{n(S)}$$
$$P(A) = \frac{33}{36}$$
$$\boldsymbol{P(A) = \frac{11}{12}}$$

(ii) Let B be the event getting the product is between 5 and 10.

$$\begin{aligned} B = \{&(1,5),(1,6),(2,3),(2,4),(2,5),(3,2),\\ &(3,3),(4,2),(5,1),(5,2),(6,1)\}\end{aligned}$$
$$n(B) = 11$$
$$P(B) = \frac{n(B)}{n(S)}$$
$$\boldsymbol{P(B) = \frac{11}{36}}$$

(iii) Let C be the event getting the difference between both the dots is equal to 4.

$$C = \{(1,5),(2,6),(5,1),(6,2)\}$$
$$n(C) = 4$$
$$P(C) = \frac{n(C)}{n(S)}$$
$$P(C) = \frac{4}{36}$$
$$\boldsymbol{P(C) = \frac{1}{9}}$$

(iv) Let D be the event getting number at least 5 on the first dice and number at least 4 on the second dice.

$$D = \{(5,4),(5,5),(5,6),(6,4),(6,5),(6,6)\}$$
$$n(D) = 6$$

$$P(D) = \frac{n(D)}{n(S)}$$
$$P(C) = \frac{6}{36}$$
$$\boldsymbol{P(C) = \frac{1}{6}}$$

3. One alphabet is selected at random from the word "$\boldsymbol{MATHEMATICS}$". Find the probability of getting: (i) vowel (ii) consonant (iii) an $\boldsymbol{E}$ (iv) an $\boldsymbol{A}$ (v) not $\boldsymbol{M}$ (vi) not $\boldsymbol{T}$

Sample space is

$$S = \{M, A, T, H, E, M, A, T, I, C, S\}$$
$$n(S) = 11$$

(i) Let A be the event getting vowel.

$$A = \{A, E, A, I\}$$
$$n(A) = 4$$
$$P(A) = \frac{n(A)}{n(S)}$$
$$\boldsymbol{P(A) = \frac{4}{11}}$$

(ii) Let B be the event getting consonant.

$$B = \{M, T, H, M, T, C, S\}$$
$$n(B) = 7$$

$$P(B)=\frac{n(B)}{n(S)}$$
$$\boldsymbol{P(B)=\frac{7}{11}}$$

(iii) Let C be an event getting of getting an E.

$$C=\{E\}$$
$$n(C)=1$$
$$P(C)=\frac{n(C)}{n(S)}$$
$$\boldsymbol{P(C)=\frac{1}{11}}$$

(iv) Let D be the event getting an A.

$$D=\{A,A\}$$
$$n(D)=2$$
$$P(D)=\frac{n(D)}{n(S)}$$
$$\boldsymbol{P(D)=\frac{2}{11}}$$

(v) Let E be the event getting M.

$$E=\{M,M\}$$
$$n(E)=2$$
$$P(E)=\frac{n(E)}{n(S)}$$

$$P(E)=\frac{2}{11}$$

To find probability not getting M, we have

$$P(E')=1-P(E)$$
$$P(E')=1-\frac{2}{11}$$
$$P(E')=\frac{11-2}{11}$$
$$\boldsymbol{P(E')=\frac{9}{11}}$$

(vi) Let F be the event getting T.

$$F=\{T,T\}$$
$$n(F)=2$$
$$P(F)=\frac{n(F)}{n(S)}$$
$$P(F)=\frac{2}{11}$$

To find probability not getting T, we have

$$P(F')=1-P(F)$$
$$P(F')=1-\frac{2}{11}$$
$$P(F')=\frac{11-2}{11}$$
$$\boldsymbol{P(F')=\frac{9}{11}}$$

4. Aslam rolled a dice. What is the probability of getting the numbers 3 or 4? Also find the probability of not getting the numbers 3 or 4.

When a dice is rolled, the sample space will be:

$$S=\{1,2,3,4,5,6\}$$
$$n(S)=6$$

Let A be an event getting the numbers 3 and 4

$$A=\{3,4\}$$
$$n(A)=2$$
$$P(A)=\frac{n(A)}{n(S)}$$
$$P(A)=\frac{2}{6}$$
$$\boldsymbol{P(A)=\frac{1}{3}}$$

To find probability not getting the numbers 3 and 4, we have

$$P(A')=1-P(A)$$
$$P(A')=1-\frac{1}{3}$$
$$P(A')=\frac{3-1}{3}$$
$$\boldsymbol{P(A')=\frac{2}{3}}$$

5. Abdul Hadi labelled cards from 1 to 30 and put them in a box. He selects a card at random. What is the probability that selected card contains:
(i) the number 25
(ii) number between 17 to 22
(iii) number at least 20
(iv) number not 27 and 29
(v) number not between 12 to 15

Sample space is

$$S=\{1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30\}$$
$$n(S)=30$$

(i) Let A be the event getting the number 25.

$$A=\{25\}$$
$$n(A)=1$$
$$P(A)=\frac{n(A)}{n(S)}$$
$$\boldsymbol{P(A)=\frac{1}{30}}$$

(ii) Let B be the event getting number between 17 to 22.

$$B=\{17,18,19,20,21,22\}$$
$$n(B)=6$$

$$P(B) = \frac{n(B)}{n(S)}$$
$$P(B) = \frac{6}{30}$$
$$\boldsymbol{P(B) = \frac{1}{5}}$$

(iii) Let C be an event of getting number at least 20.

$$C = \{20,21,22,23,24,25,26,27,28,29,30\}$$
$$n(C) = 11$$
$$P(C) = \frac{n(C)}{n(S)}$$
$$\boldsymbol{P(C) = \frac{11}{30}}$$

(iv) Let D be the event getting numbers 27 and 29.

$$D = \{27, 29\}$$
$$n(D) = 2$$
$$P(D) = \frac{n(D)}{n(S)}$$
$$P(D) = \frac{2}{30}$$
$$P(D) = \frac{1}{15}$$

To find probability not getting numbers 27 and 29, we have

$$P(D') = 1 - P(D)$$
$$P(D') = 1 - \frac{1}{15}$$
$$P(D') = \frac{15-1}{15}$$
$$\boldsymbol{P(D') = \frac{14}{15}}$$

(v) Let E be the event getting number between 12 to 15.

$$E = \{12,13,14,15\}$$
$$n(E) = 4$$
$$P(E) = \frac{n(E)}{n(S)}$$
$$P(E) = \frac{4}{30}$$
$$P(E) = \frac{2}{15}$$

To find probability not getting number between 12 to 15, we have

$$P(E') = 1 - P(E)$$
$$P(E') = 1 - \frac{2}{15}$$
$$P(E') = \frac{15-2}{15}$$
$$\boldsymbol{P(E') = \frac{13}{15}}$$

6. The probability that Ayesha will pass the examination is 0.85. What will be the probability that Ayesha will not pass the examination?

Let A be the event that Ayesha will pass the examination

$$P(A) = 0.85$$

To find probability that Ayesha will not pass the examination, we have

$$P(A') = 1 - P(A)$$
$$P(A') = 1 - 0.85$$
$$\boldsymbol{P(A') = 0.15}$$

7. Taabish tossed a fair coin and rolled a fair dice once. Find the probability of the following events:
(i) tail on coin and at least 4 on dice
(ii) head on coin and the number 2,3 on dice
(iii) head and tail on coin and the number 6 on dice
(iv) not tail on coin and the number 5 on dice
(v) not head on coin and the number 5 and 2 on dice

If tossed a fair coin and rolled a fair dice once than sample space is

$$S = \{(H,1),(H,2),(H,3),(H,4),(H,5),(H,6),(T,1),(T,2),(T,3),(T,4),(T,5),(T,6)\}$$
$$n(S) = 12$$

(i) Let A be the event getting the tail on coin and at least 4 on dice.

$$A = \{(T,4),(T,5),(T,6)\}$$
$$n(A) = 3$$
$$P(A) = \frac{n(A)}{n(S)}$$
$$P(A) = \frac{3}{12}$$
$$\boldsymbol{P(A) = \frac{1}{4}}$$

(ii) Let B be the event getting head on coin and the number 2,3 on dice.

$$B = \{(H,2),(H,3)\}$$
$$n(B) = 2$$
$$P(B) = \frac{n(B)}{n(S)}$$
$$P(B) = \frac{2}{12}$$
$$\boldsymbol{P(B) = \frac{1}{6}}$$

(iii) Let C be an event getting head and tail on coin and the number 6 on dice

$$C = \{(H,6),(T,6)\}$$
$$n(C) = 2$$
$$P(C) = \frac{n(C)}{n(S)}$$
$$P(C) = \frac{2}{12}$$
$$\boldsymbol{P(C) = \frac{1}{6}}$$

(iv) Let D be the event getting tail on coin and the number 5 on dice.

$$D = \{(H,5)\}$$
$$n(D) = 1$$
$$P(D) = \frac{n(D)}{n(S)}$$
$$P(D) = \frac{1}{12}$$

To find probability getting not tail on coin and the number 5 on dice, we have

$$P(D') = 1 - P(D)$$
$$P(D') = 1 - \frac{1}{12}$$
$$P(D') = \frac{12-1}{12}$$
$$\boldsymbol{P(D') = \frac{11}{12}}$$

(v) Let E be the event getting head on coin and the number 5 and 2 on dice.

$$E = \{(H,5),(H,2)\}$$
$$n(E) = 2$$
$$P(E) = \frac{n(E)}{n(S)}$$
$$P(E) = \frac{2}{12}$$
$$P(E) = \frac{1}{6}$$

To find probability getting not head on coin and the number 5 and 2 on dice, we have

$$P(E') = 1 - P(E)$$
$$P(E') = 1 - \frac{1}{6}$$
$$P(E') = \frac{6-1}{6}$$
$$\boldsymbol{P(E') = \frac{5}{6}}$$

8. A card is selected at random from a well shuffled pack of 52 playing cards. What will be the probability of selecting:
(i) a queen
(ii) neither a queen nor a jack

A random card is selected from 52 playing cards

$$n(S) = 52$$

(i) Let Q be the event of getting a queen, so

$$n(Q) = 4$$
$$P(Q) = \frac{n(Q)}{n(S)}$$
$$P(Q) = \frac{4}{52}$$
$$\boldsymbol{P(Q) = \frac{1}{13}}$$

(ii) Let R be the event of getting a queen or jack, so

$$n(R) = 8$$
$$P(R) = \frac{n(R)}{n(S)}$$
$$P(R) = \frac{8}{52}$$
$$\boldsymbol{P(R) = \frac{2}{13}}$$

To find probability of having neither a queen nor a jack, we have

$$P(R') = 1 - P(R)$$
$$P(R') = 1 - \frac{2}{13}$$
$$P(R') = \frac{13-2}{13}$$
$$\boldsymbol{P(R') = \frac{11}{13}}$$

9. A card is chosen at random from a pack of 52 playing cards. Find the probability of getting: (i) a jack (ii) no diamond

A random card is selected from 52 playing cards

$$n(S) = 52$$

(i) Let J be the event of getting a jack, so

$$n(J) = 4$$
$$P(J) = \frac{n(J)}{n(S)}$$
$$P(J) = \frac{4}{52}$$
$$\boldsymbol{P(J) = \frac{1}{13}}$$

(ii) Let D be the event of getting a diamond, so

$$n(D) = 13$$

$$P(D) = \frac{n(D)}{n(S)}$$
$$P(D) = \frac{13}{52}$$
$$\boldsymbol{P(D) = \frac{1}{4}}$$

To find the probability of not getting diamond, we have

$$P(D') = 1 - P(D)$$
$$P(D') = 1 - \frac{1}{4}$$
$$P(D') = \frac{4 - 1}{4}$$
$$\boldsymbol{P(D') = \frac{3}{4}}$$